

TBP Assisted Synthesis of High Concentration Quasi-Type II CdSe/CdS Dot-in-Rods Nanocrystals and its Chirality Study

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Abstract

We present a novel TBP assisted synthesis method for green emission CdSe/CdS DRs, which exhibit ligand-induced chiral activity no less than red-emitting, the CD g-factor can reach 1.3×10^{-3} , coupling with excellent CPL activity ($g_{lum} 5 \times 10^{-4}$). This achievement will lay the foundation for the realization of full-color CPL.

Author Keywords

Induced chirality, CdSe/CdS, Quasi-type II, Circularly polarized luminescence, Circular dichroism

1. Introduction

Circularly polarized luminescence (CPL) has drawn great attention due to its potential application in 3D displays^[1], CPL laser^[2] and optical sensors^[3]. We have studied the structure-activity relationship between the morphology of semiconductor nanocrystals (NCs) and chiral activity, and achieved ligand-induced CPL activity by precisely adjusting the morphology of CdSe/CdS core/shell NCs, including dot-in-rods (DRs)^[4], Tadpoles^[5] and nanoplatelets^[6]. It provides the possibility for the study of fluorescent chiral optical behavior of NCs, however, the current research on CPL is still in its infancy, which is limited in the aspect of morphology and luminescence retention of semiconductors, especially for green emission CPL. In order to achieve the high-quality displays, the green emission is as important as that of red emission, and the research of green-emitting semiconductor NCs is still the bottleneck.

In this work, we will investigate how to achieve green-emitting CPL. Srivastava, et al.^[7] and Banin, et al.^[8] demonstrate fabrication of green DRs, which further realizing DRs on-chip LEDs for display backlighting.

However, due to the core particle size of green luminescent quantum rods is too small (less than 2.4 nm)^[9], the traditional preparation process mainly relies on rapid cooling, which results in low raw material utilization, even less than 10%, and a large amount of precursors exist in the system. This increases the difficulty of preparation of green quantum rods, including the purification of unreacted precursors and control of shell thickness.

In order to solve this problem, we present a novel Tributyl phosphine (TBP) assisted synthesis method for high concentration quasi-type II CdSe/CdS DRs. Here, we innovatively added Se-TBP precursor under high-temperature reaction to improve the precursor activity, which give us the chance to achieve the preparation of high-concentration CdSe core. Compared with the traditional preparation process, the raw material utilization rate is greatly improved. Based on this method, we deeply studied the precise control of the morphology of green DRs, including aspect ratio, shell thickness, etc. And, the results show that this green-emitting quasi-type II DRs exhibit ligand-induced chiral activity no less than red-emitting, the circular dichroism (CD) g-factor can reach 1.3×10^{-3} for cysteine-modified NCs, coupling with excellent CPL activity ($g_{lum} 5 \times 10^{-4}$). In the future, we will further study how to improve the chiral properties of green CPL semiconductor NCs, and this achievement will lay the foundation for the realization of full-color 3D display.

2. Experiments

Chemicals:

Trioctylphosphine oxide (TOPO 99%), Trioctylphosphine (TOP, 97%), Tributylphosphine (TBP, 97%), Octadecylphosphonic acid (ODPA, 99%), tetradecylphosphonic acid (TDPA, 98%), and

hexylphosphonic acid (HPA, 99%) were purchased from Strem Chemicals. Cadmium oxide (CdO, 99.99%) and selenium (Se, 99.99%) were purchased from Sigma-Aldrich. Tetramethylammonium hydroxide solution (25% in Water), sulfur (S, 99.99%), L-cysteine hydrochloride monohydrate (L-Cys, 99%), and D-cysteine hydrochloride monohydrate (D-Cys, 99%) were purchased from Aladdin Industrial Co., Ltd (Shanghai, China). Pure water was purchased from Hangzhou Wahaha Group Co., Ltd, China. All chemicals were used as received without further purification.

Synthesis of the CdSe core

High concentration CdSe core were prepared by a modified procedure.^[10-12] TOPO (1.5g), ODPa (0.140g) and CdO (0.030g) were mixed in a 50mL flask, heated to 150°C and exposed to vacuum and Argon alternately over 5 times until CdO showed as brown solid and the rest showed colorless liquid. Then, under argon, the solution was heated to above 300°C to dissolve the CdO until it turned optically clear and colorless which showed that the reaction between CdO and ODPa was complete. Then the temperature was continued raising to 320°C, 1.5mL TOP was injected in the flask and the temperature was to 300°C naturally due to the injection of TOP. Then the Se:TBP solution (0.4mL, 1mol/L) was injected as the temperature of 320°C. The main difference from traditional methods is the Se-TBP precursor.

The amount of CdSe seeds was determined with absorption spectroscopy using the empirical formulas of Peng et al.^[13] which correlates the position of the first electronic transition with the diameter of the CdSe particles (equation 1) and the molar extinction coefficient of the particles and their diameter (equation 2).

$$D = 1.6122 \times 10^{-9} \cdot \lambda^4 - 2.6575 \times 10^{-6} \cdot \lambda^3 + 1.6242 \times 10^{-3} \cdot \lambda^2 - 0.4277 \cdot \lambda + 41.57 \quad (1)$$

$$\epsilon = 5857 \cdot (D)^{2.65} \quad (2)$$

Synthesis of quasi-type II CdSe/CdS DRs

The green-emitting quasi-type II CdSe/CdS DRs were synthesized via our previous “TOP-assisted SILAR” method^[14-17].

Ligand Exchange on the CdSe/CdS DRs with Chiral Cysteine Molecules

The cysteine ligand exchange reaction was carried out using a previously reported method^[4, 18, 19]. The D- or L-cysteine hydrochloride monohydrate (0.2 M) was dissolved in DI water, and then the pH of the solution was adjusted to 12 by TMAH (25% in water) to form the cysteine solution. Then, a solution of CdSe/CdS DRs in hexane (2 mL) was added to the cysteine solution (2 mL), and the reaction mixture was deoxygenated and stirred at room temperature under argon in the absence of light for 48 h. The reaction mixture was left to stand for 1 h to allow the phases to separate. The bottom aqueous layer was got with a syringe, and the Cys-CdSe/CdS DRs were purified by precipitation with ethanol/chloroform/DI water (4:1:1, 2 times). The purified Cys-DRs were dissolved in DI H₂O and stored at room temperature in the dark.

Characterization:

Transmission electron microscopy (TEM) experiments were carried out with an FEI Tecnai G2 F30 operating at 300 kV. The UV-vis absorption spectra were measured using a UV-vis spectrometer (PerkinElmer, Lambda365), and the photoluminescence (PL) spectra were recorded on a PerkinElmer LS45. The CD measurements were conducted on a JASCO J-1500 CD spectrometer at a scan rate of 100 nm/min. All CD experiments were carried out in DI water with a quartz cuvette (1 cm path length). The CPL measurements were performed on a JASCO CPL-300 spectrometer in DI water with a quartz cuvette (1 cm path length) with an excitation wavelength of 405 nm. All optical measurements were performed at room temperature under ambient conditions.

3. Results and discussion

In this work, A high-yield preparation process based on TBP assisted synthesis method has been discovered, and based on this, we have achieved high concentration CdSe core with UV absorption of 489 nm. Diameter of the CdSe core is determined from the absorption spectrum by Peng's method (equation 1)^[13], it can be calculated that the particle size is 2.2 nm and the concentration reaches 7 mmol/L. Compared with traditional processes, the core concentration can be increased by dozens of times. This exciting result is based on the heat injection temperature far beyond the boiling point of TBP, which may be due to this specific state producing this beneficial effect.

We have successfully synthesized green-emitting quasi-type II CdSe/CdS DRs via our previous “TOP-assisted SILAR” method^[14-17]. Different from Type I, the shell of quasi-Type II quantum rods cannot perfectly confine excitons and have a greater possibility of achieving high chirality transfer effects (Fig. 1).

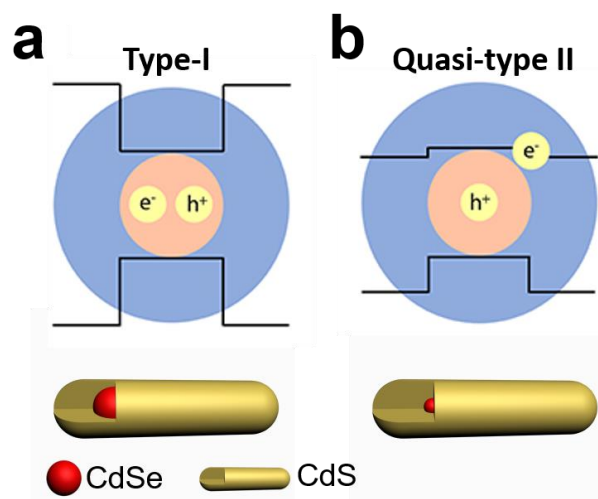


Figure 1. Schematic diagram of valence band in type I (a) and quasi-type II (b).

Due to the high utilization rate of precursors in the core, unreacted or incompletely reacted precursors will not have a major impact on quantum rod growth during the shell growth process. Therefore, we can achieve a quantum rod with a basically

unchanged shell but a changing aspect ratio by simply adjusting the reaction time or the concentration of the core. As shown in Fig. 2, by changing the reaction time, quasi-type II green-emitting DRs with different aspect ratios can be obtained. For instance, AR=2.6 is obtained in 1 min, AR=3.4 is obtained in 2 minutes, and AR=4.8 is obtained in 4 minutes.

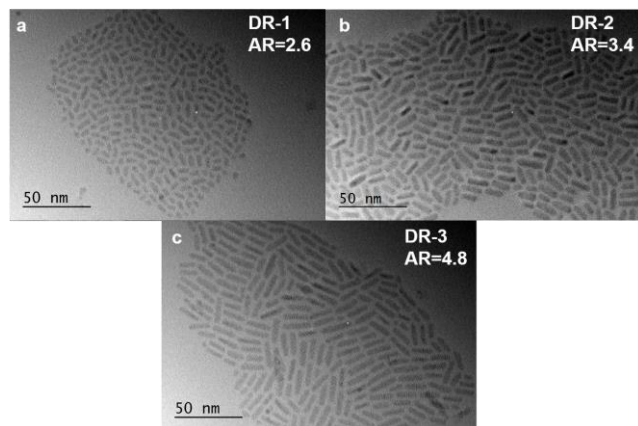


Figure 2. TEM images of quasi-type II CdSe/CdS DRs with different aspect ratios (AR). Different AR can be obtained by various reaction times, (a) 1min, (b) 2min, (c) 4min.

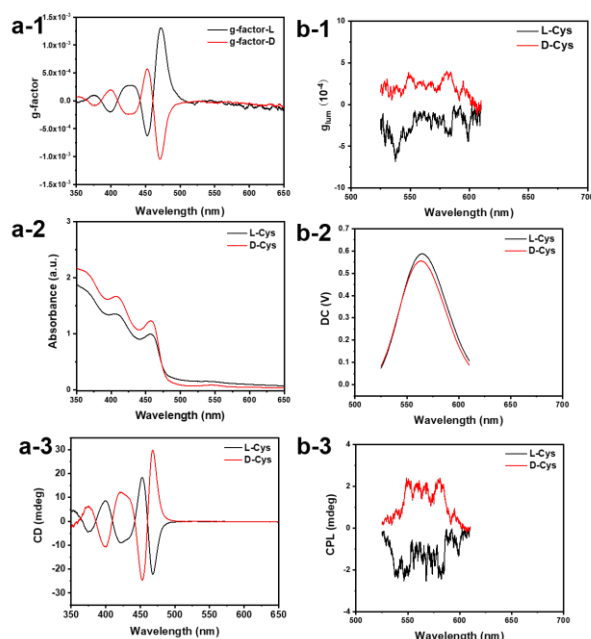


Figure 3. Chirality study of quasi-type II CdSe/CdS DRs (a) CD spectra evolution of DRs-3 with aspect ratio of 4.8, (b) the corresponding CPL spectra.

Fig. 3 present the CD and CPL spectra for DR-3 with aspect ratio of 4.8. In the vicinity of the exciton absorption, mirrored CD line shapes are presented when L- and D-cysteine are independently used as surface ligands. The anisotropic g-factors were calculated to quantitatively compare the CD responses of these DRs, and the maximum CD g-factor can reach 1.3×10^{-3} at 472.4 nm, which

exhibit ligand-induced chiral activity no less than red-emitting^[17, 20]. Moreover, the CPL activities can be observed because of the special anisotropic morphology (Fig. 2), with the highest g_{lum} value of 5×10^{-4} at 555 nm for DR-3.

4. Conclusion

In summary, we have described detailed synthetic methods for quasi-type II CdSe/CdS DRs. Due to the introduction of TBP, high concentration implementation has become possible, and we have also achieved DRs with different aspect ratios (AR), such as luminescence below 560 nm from 2.6 to 4.8. The maximum CD g-factor can reach 1.3×10^{-3} at 472.4 nm, coupling with excellent CPL activity ($g_{lum} 5 \times 10^{-4}$ at 555 nm). This TBP assisted synthesis method provides a strong advantage for delicate control of optical properties, especially in the preparation of small-sized cores (less than 2.4 nm), as it can consume core precursors as much as possible and reduce their impact on the shell growth process. In principle, this method allows composite fabrications using QDs covering a wide range of sizes and chemical nature. And the as prepared green CPL semiconductor NCs have great potential to be used in full-color 3D display.

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